

ViFiCon: Vision and Wireless Association Via Self-Supervised Contrastive Learning

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Abstract

We introduce ViFiCon, a self-supervised contrastive scheme which learns a cross-modal association between vision and wireless modalities. Specifically, the system uses pedestrian data collected from RGB-D camera footage and WiFi Fine Time Measurements (FTM) from a user’s smartphone device. Depth data from RGB-D (vision domain) is inherently linked with an observable pedestrian, but FTM data (wireless domain) is associated only to a smartphone on the network. We represent temporal sequences from both vision and wireless domains by stacking multi-person depth data sequences within an image representation. This simplicity allows both scene-wide processing and fewer vision and wireless features, alleviating privacy and energy associated with transmitting IMU data. To facilitate self-supervised learning, we design a scene-wide synchronization pretext task for our network and then employ the learned representation for the downstream multimodal association task. We show that compared to fully supervised state-of-the-art models, ViFiCon achieves high performance vision-to-wireless association of **92.63%** in 25 frames sliding window fashion (2.5s), finding which bounding box corresponds to which smartphone device, without hand-labeled association examples for training data. Extensive experimental results demonstrate ViFiCon applicability in real-world systems when wireless data annotations are scarce.

1. Introduction

With the proliferation of sensors from multiple modalities, cross-domain association for vision and wireless has become a fundamental task in many real-world systems [6, 9, 31, 55]. The association challenge, i.e. correctly matching network addresses with the entities detected in camera images, enables communication applications such as alerting pedestrians to nearby danger conditions including impassable sidewalks, unusual crowding, high-risk crosswalks, or construction zones. Additionally, this asso-

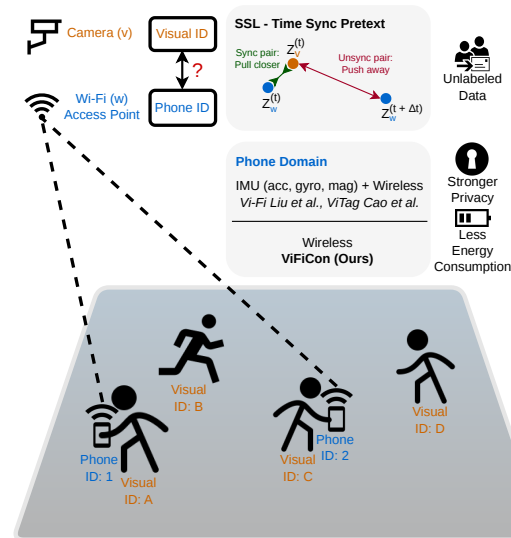


Figure 1. Associating visual observations to WiFi sequences enables novel applications where observed users can be linked to their smart phones. Our approach to learning vision-to-wireless associations does not require annotated ground-truth matches. Instead, a pretext task of temporal synchronization guides multimodal representation learning for the downstream task of associating individual bounding boxes to specific smartphones.

ciation supports vision applications in low-light or highly cluttered environments where wireless signals can be used to improve camera-image detection.

Existing approaches associating information across modalities use visual features with laser ranging, motion/gait patterns or thermal imagery [25, 40, 51, 55]. Among multimodal methods for vision and wireless, RGB-W [4] fuses camera and received sequence strength (RSS), and [16, 64] fuse camera and WiFi channel state information (CSI). Prior vision and wireless methods are limited by requiring multiple access points (AP). Single AP methods for cross-model association of vision and WiFi has emerged in recent state-of-the-art work, including Vi-Fi [31] and

ViTag [6]. Our approach *ViFiCon* improves efficiency by using fewer features overall from both RGB-D and WiFi Fine Time Measurements (FTM), relying only on temporal depth data. Moreover, unlike Vi-Fi and ViTag, *ViFiCon* requires no hand-labeled data, improving its practicality for real-world applications.

Challenges. To facilitate vision-phone association to a wider range of applications in real world, existing systems encounter several new challenges: *1) Annotations.* Prior works [6, 31] rely heavily on ground truth labels in order to train their models in a supervised learning manner. However, time-consuming and labor-expensive annotation hinders the adoption of these systems in novel environments, especially for wireless data which cannot be hand labeled using human visual perception; *2) Feature abundance.* In order to achieve high association accuracy, previous methods consider a wealth of features in phones including accelerations, gyroscope and magnetometer readings as well as wireless data jointly. This assumption can break in a real system since a user may disable capture and transmission of these data due to privacy [45] or energy concerns [22].

Approach. To address these challenges, we present *ViFiCon*, a self-supervised contrastive learning model to associate vision and wireless data, circumventing the need of hand-crafted labels. We demonstrate our approach’s practical association capability in using only wireless data, alleviating the use of inertial sensors from prior works.

ViFiCon is among the *first* works to use a shared representation for both vision and wireless modalities using a Siamese model structure. Multimodal associations are obtained without hand-labeled data. We construct positive and negative pairings across the two modalities through passively collected timestamp information, as inspired by temporal contrastive audio and video synchronization in [23]. We create a novel representation of sequences from the two modalities. Combining a set of sequences in one shared image learns a global scene-wide synchronization, where each person’s FTM and depth in the scene is shown for a window. This guides the downstream one-to-one association task, where we represent each person’s depth and FTM data in their own images for sequence matching. We train a Siamese convolutional neural network model on the scene-wide synchronization task to embed positive and negative pairs into a joint latent space with a Euclidean distance contrastive loss. We can then apply the task downstream without any more training. We show the motivation of *ViFiCon* in Figure 1.

Summary of Contributions We summarize our contributions as follows:

- We introduce a novel representation of depth and FTM data coined *Vision Depth Sequence (VDS)* and *Wireless Depth Sequence (WDS)* for scene-wide representation without IMU data from phone domain.

- We craft a self-supervised learning pretext synchronization task for vision and wireless, using passively collected timestamp information to learn a multimodal joint embedding space without human-labeled data.
- *ViFiCon* achieves an **92.63%** Identity Precision (IDP) in 25-frame temporal window view (2.5s), competitive with fully-supervised approaches while using a simplistic design for a real application.

2. Related Work

Multimodal Association Multimodal association enhances a model’s performance by providing additional scene context from shared knowledge across domains. The modalities are fused by mapping data into a shared latent space, where the representations can be compared. Multimodal association has been applied domains such as audio and video [11, 23, 26, 63], audio and text [18], and video and wireless [38, 39, 64]. We build on prior work that associates vision with wireless sensor data. Wireless data complements visual sensors by overcoming obstructions [3] and compensating when visual frames are unavailable [8, 60–62, 64], hence we leverage WiFi signals with vision to re-identify and tracking in such scenarios. RGB-W [4] leverages video with RSS from phones to associate boxes with MAC addresses indoors, but requires multiple AP. WiVi and Eye-Fi [16, 64] use CSI along with vision, however, performance degrades under poor lighting and wireless signal interference, and may require multiple AP. Other techniques use a fusion of FTM, IMU, and vision to perform association and tracking between domains while using one AP and an affinity matrix [31, 32], the X-Translator network [6, 7] or a GAN-based approach [33]. We simplify the multimodal association by using vision and wireless sequences mapped into a common space.

Self-Supervised Learning Learning without ground truth labels via Self-Supervised Learning (SSL) has been studied in many domains. The core intuition is to learn meaningful features from unlabeled data, usually by a set of pretext tasks. Examples include reconstructing by masking tasks to learn the context, or a similarity latent space by contrastive learning. These tasks have been applied successfully in a wide range of applications in different realms, including learning unimodality in vision [10, 19, 21, 44, 56], NLP [13], audio [37], inertial sensors [57] and wireless [5, 28, 28, 30, 58, 59] domains. Though joint domains have been studied, they are mainly limited in vision and language [12, 27, 49]. In comparison, our work explores the unique research question for real-world multimodal association systems by leveraging contrastive learning to bridge **vision** and **wireless** features jointly.

WiFi Fine Time Measurements (FTM) The FTM protocol introduced in the IEEE 802.11-2016 Standard [2, 35] allows for distance with a margin of error to be collected indoor

and outdoor through wireless ranging. Round trip time is calculated between the AP and the connected device, with the range then estimated using propagation speed to bring meter-level precision [20].

3. System Overview

3.1. Dataset

Several existing datasets pose limitations to the task of vision-phone association, due to their uni-modality limit such as ImageNet [48], COCO [29], MOT [41] for visual tasks in the vision domain, or IMU data (e.g. HHAR [52], UCI [46], Motion Sensen [36], Shoaib [50]) for the human activity recognition (HAR) task [36, 52] in the phone domain. In comparison, *Vi-Fi* [15] provides a **unique joint** dataset that combines various modalities including vision and phone with IMU and WiFi FTM readings. This comprehensive multimodality offers us a suitable dataset and therefore is chosen for our experiments. We present the necessity of dataset information in this paper and refer readers to the details of data collection in *Vi-Fi* [31].

From the *Vi-Fi* dataset, sequences in outdoor scenarios are used, including 3 participants and a number of passers-by in the background with at most 11 persons in one frame.

3.2. Data Pre-Processing

We formally present the details of data pre-processing in our approach.

Camera Data. We follow the main procedure described in *ViTag* [6] to tracklets, which are a sequence of one modality’s data. We use a time series sequence of bounding boxes (*BBX* to represent a Tracklet from camera data (T_c)). A bounding box is defined as: $BBX = [x, y, d, w, h]$, $T_c \in \mathbb{R}^{k \times 5}$, where the centroid of the bounding box is represented by x and y while d depth measurement. Bounding box sizes are represented by the width w and height h , respectively. We use k to indicate the window’s length – the number of datapoints in the temporal sequence in a window. Unlike prior works [6, 31] using all five dimensional camera data, we represent a Tracklet $T_v \in \mathbb{R}^{k \times 1}$ in a much simpler form that utilizes only depth information: A subject’s depth in one frame is summarized as a scalar by averaging all the depth values of the region within its bounding box.

Wireless Data. We use the range of FTM measurements to form the sequence of wireless data, denoted as a Tracklet $T_w \in \mathbb{R}^{k \times 1}$.

Multimodal Synchronization. Due to various sampling rate in the raw data, we perform synchronization of all the modalities before feeding to the model. Network Time Protocol (NTP) on the devices is used to synchronize the camera and phone data. The camera frame original sampling rate is 30 FPS, while FTM is sampled by a much lower rate at 3-5 Hz. We therefore sub-sample the depth data with 10

FPS to better capture the movement of pedestrians and participants in the scene. To upsample FTM, different from *ViTag* [6] where a wireless datapoint finds a camera frame with the closest timestamp, we extract the k nearest larger and smaller timestamps from the FTM data and perform a linear interpolation to align the sequences. This allows better data for the missing FTM datapoints.

3.3. Vision & Wireless Depth Sequence Representation

The depth from the RGB-D camera and FTM information for an individual is a 1-dimensional sequence of values varying over each timestamp. Prior work either deals with a fixed number of pedestrians in a scene a [31], or processes each tracklet individually [6], leaving the burden of associating all samples in a scene to an association module. These representation designs pose some limitations that cannot handle association when the number of correspondences N_c exceeds the predefined parameter, or the complexity scales linearly by N_c . To push these limits, we here introduce novel representations coined **Vision Depth Sequence (VDS)** and **Wireless Depth Sequence (WDS)**, formally defined as $D_v \in \mathbb{R}^{H \times W}$ and $D_w \in \mathbb{R}^{H \times W}$ where H and W is the resolution of the image. Specifically, a Tracklet (e.g. T_v) is mapped to a unique set of gray-scale values in one dimension for each frame that forms a row. We expand the row with some height values represented in one row in a D_v image. When multiple samples are present in scene, we stack all their corresponding representations vertically separated by gray padding in rows whose color is determined by the average of the two to clearly define the sequences.

This simple depth image representation enjoys two advantages: First, it allows multiple sequences of walking data to be processed in the same frame, resulting in a global activity view for the scene. Second, finding the number of correspondences N_c between the depth and FTM data is trivial in D . We summarize the characteristics in comparison to prior works in Table 1.

Method	No Fixed N_c	Granularity	Complexity
<i>Vi-Fi</i>	×	Scene	O(1)
<i>ViTag</i>	✓	Subject	$O(N_c)$
<i>ViFiCon</i>	✓	Scene	O(1)

Table 1. Summary of the characteristics of representations. Compared to prior works, Vision Depth Sequence (VDS) and Wireless Depth Sequence (WDS) enjoy the features of both no fixed number of correspondences N_c and scene-wise computation, yielding time complexity of processing one scene in $O(1)$.

This representation combines the benefit from both the

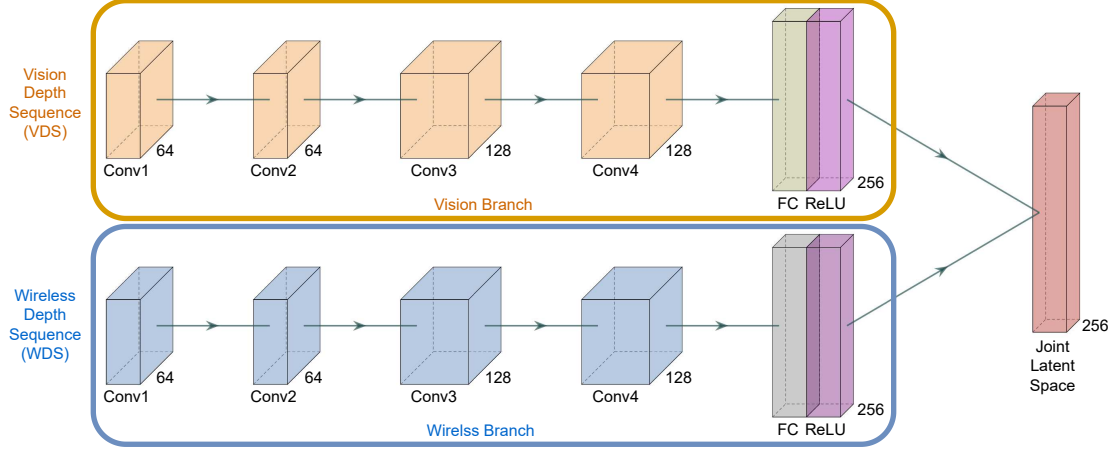


Figure 2. Network Architecture. Inspired by SimCLR [10], we employ a dual-stream convolutional neural network structure to process the representations of the vision and wireless modalities. Each branch learns independent domain features followed by a shared latent space. Network updates are performed through the contrastive loss. We show that this simplicity is sufficient and beneficial in real world scenarios in terms of efficiency.

existing work of *Vi-Fi* [31] and the model *X-Translator* in *ViTag* [6]. To be specific, *Vi-Fi* handles a fixed number of pedestrians for association (N_c), while *ViFiCon* and *X-Translator* do not pose limits on N_c . However, *X-Translator* processes data for each subject in sequence while *ViFiCon* and *Vi-Fi* deal with all subjects in one scene simultaneously, resulting in the time complexity of $O(N_c)$ for the former and $O(1)$ for the latter two, respectively.

VDS & WDS Preparation We first split the multimodal data from each sequence into two separate processing steps for depth and FTM ranging data. Using an off-the-shelf ZED detection model, we extract boxes with ID values corresponding to individuals in the scene. We extract depth sequences by taking the average depth across the bounding box for each subject in one frame. The number of returned sequences varies based on the number of individuals in the frames of a window. For wireless, we extract the fixed number of ranging data for each phone. Each sequence is mapped into the Wireless Depth Sequence (WDS) for the pretext task of scene synchronization which will be described in the following sections. In the downstream association task, the sequences are split individually into their own depth sequences.

We note that the order of the sequences does not matter, as the model will pick up on spatial cues to perform matching. In the depth sequence image representation, a variable number of depth sequences may exist in the k frame temporal window. The wireless modality gives an indication at how many users in the scene hold phones, and we use this information for vision to create combinatorics of depth data. One challenge that arises is the fact that some combinatorics may not strictly match the data collected from the

wireless domain, including when only participant walking data is selected. We threshold for depth values that are not constant for a long period in the window, signifying that an individual has yet to entered or exited the scene.

4. Methodology

4.1. Pretext Task

Since each datapoint is associated with a timestamp, we consider positive pairs as temporally synchronized Vision and Wireless Depth Sequences, while negative pairs are temporally unsynchronized. Note the label in the association task is the correspondence of a pair of samples between vision and wireless domains (e.g. indicating whether they share the same ID). The main intuition of the pretext task is to leverage timestamps in the dataset, which actually act as **free** correspondence labels. This way, we can exploit data without human annotations for training. The workflow to create positive and negative pairs is outlined in Figure 3. Specifically, for a window with size k , we generate both VDS and WDS from vision and wireless data, respectively. We take two windows starting at $t - k$ and $t + k$ for example, from each we have $VDS^{(t-k)}$, $WDS^{(t-k)}$, $VDS^{(t+k)}$ and $WDS^{(t+k)}$ shown in Figure 3. Since $VDS^{(t-k)}$, $WDS^{(t-k)}$ are in the same window (sync), the contrastive loss (which will be described in later sections) will pull their latent vectors to be closer. In contrast, $VDS^{(t-k)}$ and $WDS^{(t+k)}$ are in different windows (unsync), therefore our model will push them far apart in the joint latent space. After training by the time synchronized pretext task, our model will learn a good cross-modal similarity space for the downstream association task.

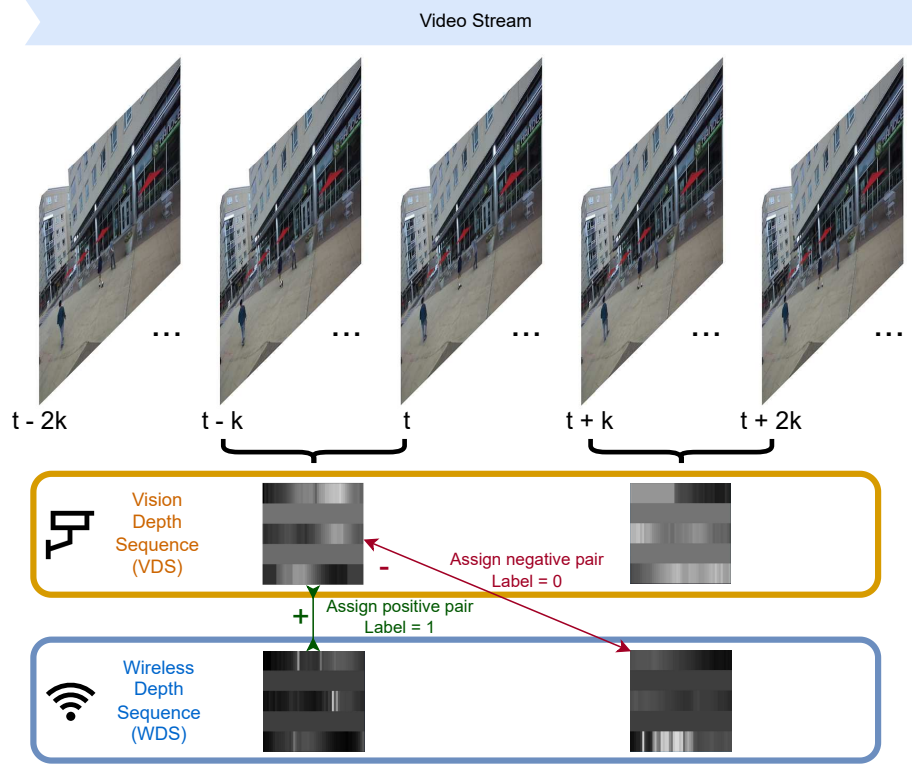


Figure 3. Self-Supervised Temporal Alignment Pretext Task. For each sequence, we use an off-the-shelf object detector to obtain bounding box information for pedestrians in the scene. We then take the average depth of the bounding box to extract depth sequences for each pedestrian, and create combinations of depth sequences with respect to the fixed number of FTM sequences in the scene if possible. Each image has depth sequences for multiple people in the scene. Temporal alignment or synchronization of these images (pre-text task) can be self-supervised and the resulting representation that can then be used for the association task (downstream task). (Best viewed when zoomed).

4.2. Model Architecture

Real-world application deployment poses challenges in acquiring computation resources often limited by end users’ budget. Albeit the success of Vision Transformer [14, 34, 42] in learning discriminative representations in recent years, this often comes at the cost of high-end computation power and large datasets [43, 53]. In addition, exploring various architectures (such as CNN vs. RNN vs. transformer) can be a promising future work. However, we focus on solving real-world problems towards applications by contrastive learning, and we show that a CNN-backbone is sufficient and efficient to achieve a high association performance using a consumer computer shown in Section 5. A summary of the differences is shown in Table 2. Our network design is inspired by SimCLR [10].

4.3. Loss Function

We create our self-supervised contrastive learning model as a set of two identical subnetworks to process the Vision and Wireless Depth Sequence representations. The result

Model	FLOPs (G) ↓	#Params (M) ↓
ViT-B	33.03	86.86
ViT-L	116.68	304.72
<i>ViFiCon</i>	0.75	25.70

Table 2. Comparison between Transformer (ViT) and CNN (*ViFiCon*) as backbones. *ViFiCon* enjoys the efficiency advantage for a real application.

is a Dual-stream CNN responsible for fusing the vision and wireless data together in a joint latent space, shown in Figure 2. Determining the embedding distance of each of the vectors in the latent space is performed using the Euclidean distance (ED), given by equation 1:

$$ED(Z_v, Z_w) = \|Z_v - Z_w\|_2 \quad (1)$$

where Z_v represents the 256-dimensional feature vector embedding $Z_v \in \mathbb{R}^{256}$ for the vision depth data, and Z_w

represents the 256-dimensional feature vector embedding $Z_w \in \mathbb{R}^{256}$ for the wireless FTM data. The loss back-propagated through the model is a standard contrastive loss function using the Euclidean distances of the feature vector embeddings. The contrastive loss (CL) is defined by equation 2:

$$CL = \frac{1}{B} \times (Y \times \|Z_v - Z_w\|_2^2 + (1 - Y) \times \max(M - \|Z_v - Z_w\|_2^2, 0)) \quad (2)$$

where $Y \in \{0, 1\}$ is the synchronization label for the given pair, M is a margin hyperparameter, and N is the number of samples. In practice we refer to batch size. For positive pairs, a small Euclidean distance value will yield to a small loss, whereas for negative pairs, so long as the Euclidean distance is larger than the margin hyperparameter M , the loss will be minimized.

We train the *ViFiCon* model on a set of 30, 3 minute outdoor sequences from the Vi-Fi using the VDS and WDS for global scene synchronization. Each depth sequence consists of a maximum of 3 sequences on both the vision side and wireless side. On the vision side, training is performed using raw bounding box data extracted from the object detector with no hand tweaking and no ground truth information. We test the model 5 different 3 minute outdoor sequences and compare with state-of-the-art fully supervised models [6, 31]. We report results of the pretext synchronization task on the test set and the generalization to the downstream association task. *ViFiCon* is trained for a total of 400 epochs at a learning rate of 0.003 and a margin hyperparameter of 0.2.

5. Experiments and Evaluation Setup

5.1. Evaluation Metrics

Following the evaluation protocol in *Vi-Fi* [31] and *ViTag* [6], Our primary evaluation metric is the IDentification Precision [47], or *IDP*. *IDP* is defined in equation 3:

$$IDP = \frac{IDTP}{IDTP + IDFP} \quad (3)$$

where *IDTP* is the identified true positive pairs, and *IDFP* is the false positive pairs in the latent space. We also provide supplementary metrics of the accuracy and F1 score for evaluating the *ViFiCon* model on both the pretext synchronization task and downstream association task

After the similarity space has been learned, it will be used for the downstream task of association depicted in Figure 4. Determining which pairs in the latent space correspond to true positives, false positives, true negatives, and false negatives is done using a margin line. We plot each of the Depth-FTM pairing Euclidean Distances in a latent space representation and then determine the best separation of positive and negative pairings. True positives are located

to the left of the margin line, and false positives are to the right. A sample view of the latent embedding space including margin line separation is shown in Figure 5.

5.2. Overall Performance

ViFiCon achieves an overall association performance with *IDP* 92.63% in the 25-frame (10FPS) sliding window on-line fashion. The high association with a short window duration of 2.5 seconds demonstrates our system’s practical usage in real-world scenarios.

Performance on Downstream Association Task We summarize the results of *ViFiCon*’s downstream association task performance on the test set in Table 3. We see that the model’s performance progressively improves in terms of accuracy (*Acc*) and *F1* score as the temporal window size increases, the best *IDP* score is obtained at a temporal window size of 25 frames.

Metrics	10F	25F	50F
<i>IDP</i> (%)	84.77	92.63	85.33
<i>Acc</i> (%)	58.33	72.65	78.23
<i>F1</i> (%)	67.10	77.38	80.00

Table 3. Summary of SSL *ViFiCon* performance on Downstream Association Task on Test Set.

Comparison to SOTA models. We compare our *IDP* score with the *Vi-Fi* [31] and *ViTag* [6] fully supervised deep learning models on the association task. Since only results on 10-frame windows are reported in *ViTag*, for a fair comparison we hereby use the same $k = 10$ for all methods albeit the highest score of *IDP* 92.63% with $k = 25$ for *ViFiCon*. We also include the hand-crafted method of pedestrian dead reckoning (PDR) and procrustes analysis (PA) [17, 24, 54]. We show that by extracting information from the vision domain to obtain the depth sequences, we are able to obtain comparable results without the need for any hand labeling of the dataset. We compare our results from temporal window size 10 with the fully supervised models on the test dataset in Table 4.

5.3. System Analysis

Performance on Pretext Synchronization Task We record the results of *ViFiCon*’s pretext synchronization task performance on the test set in Table 5. We see that in general, *ViFiCon*’s performance improves for all three metrics as the temporal window size increases, with the best results at a large number of frames as more context is provided.

Case Study: Comparing Off-The-Shelf and Manually Corrected Bounding Boxes We compare our approach with off-the-shelf detections to two different approaches. In the first, we use the ground truth participant bounding

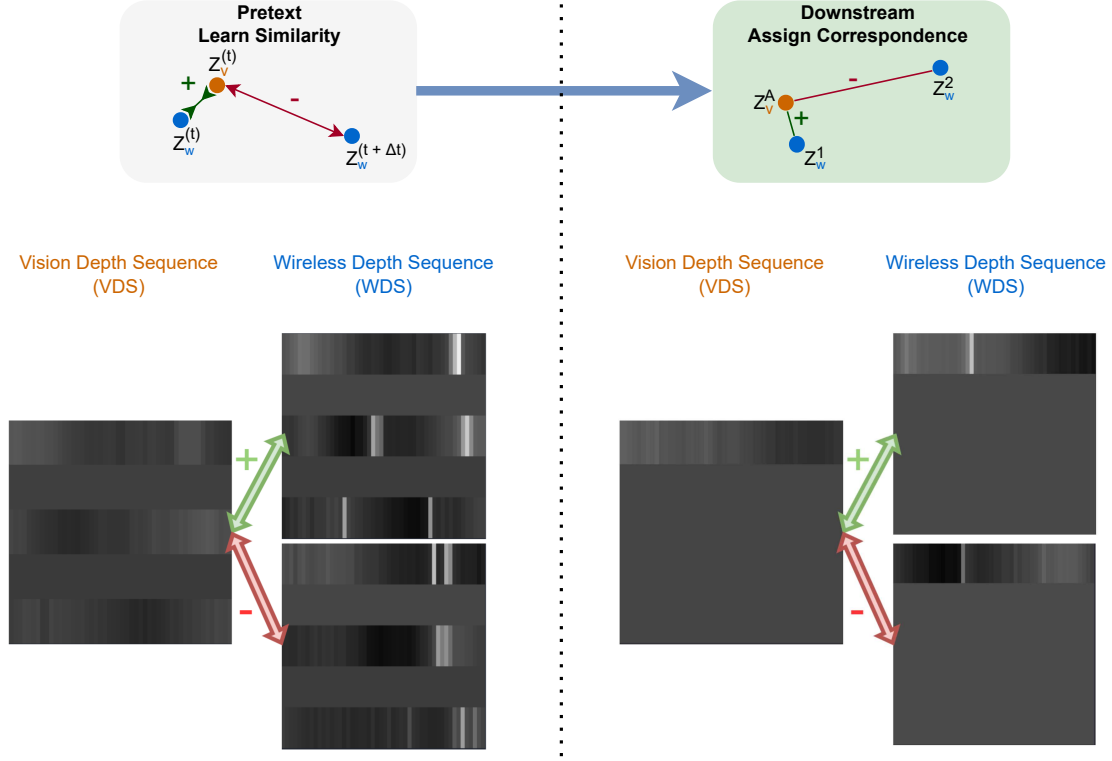


Figure 4. Global scene synchronization (frame-matching) for pretext task and individual association for downstream task. (a) The *Vision Depth Sequence (VDS)* for one frame demonstrates multiple sequences of bounding box depth values (left) and *Wireless Depth Sequence (WDS)* with FTM values (right) for pedestrians in the scene. The pretext global synchronization is a frame-matching problem taking advantage of timestamps without annotations. (b) The downstream association task is an association of a particular *VDS* to a particular FTM *WDS*, using the joint latent space learned from pretext in (a).

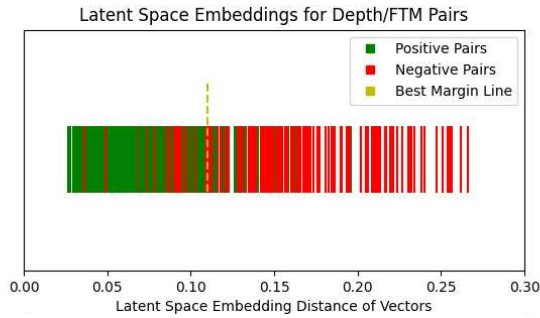


Figure 5. Sample Latent Space Embedding on downstream association task on test set. We use the true and false positive and negatives to evaluate the model performance. Observe that positive pairings of depth and FTM data (synchronized timestamps) are embedded close, whereas negative pairings (unsynchronized timestamps) have a larger distance.

boxes for the depth sequence representation on the depth side. In the second, we manually correct bounding boxes produced by the off-the-shelf model such that the IDs are

Model	<i>PDR+PA</i>	<i>Vi-Fi</i>	<i>ViTag</i>	<i>ViFiCon</i>
Training Set	-	Labeled	Labeled	Unlabeled
Features	BBX, IMU, FTM	BBX, IMU, FTM	BBX, IMU, FTM	Depth, FTM
<i>IDP</i>	41.79 %	84.97 %	87.85 %	84.77%

Table 4. Comparison of SSL *ViFiCon* performance with state-of-the-art on Downstream Association Task on Test Set as an end use system. *ViFiCon* achieves a comparable performance to SOTA deep learning models, but without ground truth annotations as well as fewer features, only requiring depth and FTM data whereas *Vi-Fi* and *ViTag* use IMU, FTM, and bounding boxes (BBX) from the camera data.

consistent by hand-tweaking. This approach ensures that a depth sequence representation containing all participants holding phones exists in the set of possible combinatorics, but does not guarantee this image will be selected in the dataset generation and train using this image. We provide tabular results of the study on the downstream task in Ta-

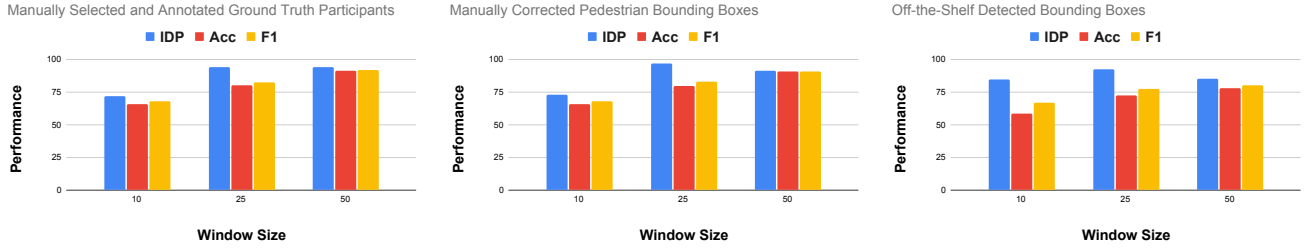


Figure 6. Metrics on manually selected and annotated ground truth (GT), manually corrected pedestrian bounding boxes, and off-the-shelf detector variants for depth data of the *ViFiCon* model on the Test Set. In general the manually annotated and selected ground truth has the best performance, though the manually corrected pedestrian bounding boxes and off-the-shelf detector perform comparably due to our thresholding technique.

Metrics	10F	25F	50F
<i>IDP</i> (%)	88.26	82.29	95.83
<i>Acc</i> (%)	71.04	78.65	85.42
<i>F1</i> (%)	75.53	79.40	86.79

Table 5. Summary of SSL *ViFiCon* performance on Pretext Synchronization Task on Test Set.

ble 6 for $k = 50$ frames, and provide a chart of the IDP for all scenarios across all temporal window sizes in Figure 6.

Method (50 Frames)	Manually Selected GT	Manually Corrected Pedestrian Bboxes	Off-the-Shelf BBoxes [1]
<i>IDP</i> (%)	94.00	91.33	85.33
<i>Acc</i> (%)	91.50	90.48	78.23
<i>F1</i> (%)	91.86	90.73	80.00

Table 6. Summary of *ViFiCon* performance on Downstream Association Task on Test Set across manually selected and annotated ground truth (GT) participants, manually corrected pedestrian bounding boxes, and off-the-shelf detector variants for depth data with temporal window size $k = 50$.

6. Conclusion

In this work, we propose *ViFiCon*, a self-supervised contrastive learning approach for vision–wireless association. *ViFiCon* leverages **unlabeled data** and requires substantially fewer phone-side features than existing SOTA methods, while maintaining competitive performance. The former is achieved by learning a visual and wireless similarity space by a pretext synchronization task in a dual-stream CNN backbone, while the latter is enabled by our novel depth sequence image representation that represents a set of depth and wireless sequences in a shared and compa-

table space. *ViFiCon* achieves an *IDP* of **92.63%** using a 25-frame (2.5s) online sliding window with only depth and FTM signals on the real-world Vi-Fi dataset, whereas fully supervised methods require per-frame correspondence labels and additional signals such as bounding boxes and IMU data.

Real-world systems poses challenges of scarce ground truth labels, privacy and energy concerns with continuous IMU sensing. We demonstrate that our self-supervised paradigm, combined with the simple yet expressive depth and wireless representations in *ViFiCon*, addresses labeling scarcity, privacy risks, and energy constraints in a unified framework. By removing reliance on identity annotations, limiting exposure to sensitive visual information, and avoiding continuous IMU usage, *ViFiCon* facilitates practical and scalable multimodal deployment, advancing vision–wireless systems toward real-world applications.

7. Limitations and Discussions

While *ViFiCon* achieves strong performance using only depth and WiFi FTM signals, several limitations remain. Our method relies on accurate pedestrian detection and depth estimation from RGB-D sensors, where detection or depth errors may degrade association quality. In addition, the approach assumes reliable temporal synchronization between camera and wireless devices, and clock drift or missing wireless measurements may affect the synchronization-based pretext task. Future directions also include exploring more expressive multimodal architectures to further improve cross-modal representation learning and association performance.

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References

- [1] Zed camera website (accessed: 04.09.2026).
- [2] "IEEE Standard for Information technology–Telecommunications and information exchange between systems Local and metropolitan area networks–Specific requirements - Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications". *IEEE Std 802.11-2016 (Revision of IEEE Std 802.11-2012)*, pages 1–3534, 2016.
- [3] Fadel Adib and Dina Katabi. See through walls with wifi! In *Proceedings of the ACM SIGCOMM 2013 conference on SIGCOMM*, pages 75–86, 2013.
- [4] Alexandre Alahi, Albert Haque, and Li Fei-Fei. Rgb-w: When vision meets wireless. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 3289–3297, 2015.
- [5] Mohammed Alloulah, Akash Deep Singh, and Maximilian Arnold. Self-supervised radio-visual representation learning for 6g sensing. In *ICC 2022-IEEE International Conference on Communications*, pages 1955–1961. IEEE, 2022.
- [6] Bryan Bo Cao, Abrar Alali, Hansi Liu, Nicholas Meegan, Marco Gruteser, Kristin Dana, Ashwin Ashok, and Shubham Jain. Vitag: Online wifi fine time measurements aided vision-motion identity association in multi-person environments.
- [7] Bryan B Cao, Abrar Alali, Hansi Liu, Nicholas Meegan, Marco Gruteser, Kristin Dana, Ashwin Ashok, and Shubham Jain. Tagging vision with smartphone identities by vision2phone translation. In *IEEE International Conference on Sensing, Communication, and Networking September 2022*, 2022.
- [8] Bryan Bo Cao, Abrar Alali, Hansi Liu, Nicholas Meegan, Marco Gruteser, Kristin Dana, Ashwin Ashok, and Shubham Jain. Vifit: Reconstructing vision trajectories from imu and wi-fi fine time measurements. In *Proceedings of the 3rd ACM MobiCom Workshop on Integrated Sensing and Communications Systems*, pages 13–18, 2023.
- [9] Siyuan Cao and He Wang. Enabling public cameras to talk to the public. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2(2):1–20, 2018.
- [10] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR, 2020.
- [11] Joon Son Chung and Andrew Zisserman. Out of time: automated lip sync in the wild. In *Asian conference on computer vision*, pages 251–263. Springer, 2016.
- [12] Bo Dai and Dahua Lin. Contrastive learning for image captioning. *Advances in Neural Information Processing Systems*, 30, 2017.
- [13] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- [14] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [15] Liu et al. Vi-fi multi-modal dataset.
- [16] Shiwei Fang, Tamzeed Islam, Sirajum Munir, and Shahriar Nirjon. Eyefi: Fast human identification through vision and wifi-based trajectory matching. In *2020 16th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, pages 59–68. IEEE, 2020.
- [17] John C Gower. Generalized procrustes analysis. *Psychometrika*, 40(1):33–51, 1975.
- [18] Yue Gu, Xinyu Li, Shuhong Chen, Jianyu Zhang, and Ivan Marsic. Speech intention classification with multimodal deep learning. In *Canadian conference on artificial intelligence*, pages 260–271. Springer, 2017.
- [19] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16000–16009, 2022.
- [20] Mohamed Ibrahim, Hansi Liu, Minitha Jawahar, Viet Nguyen, Marco Gruteser, Richard Howard, Bo Yu, and Fan Bai. Verification: Accuracy evaluation of wifi fine time measurements on an open platform. In *Proceedings of the 24th Annual International Conference on Mobile Computing and Networking*, pages 417–427. ACM, 2018.
- [21] Longlong Jing, Ling Zhang, and Yingli Tian. Self-supervised feature learning by cross-modality and cross-view correspondences. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1581–1591, 2021.
- [22] Supriya Kalyankar-Narwade, Ramesh Kumar Chidambaram, and Sanjay Patil. Estimation of parallel hybrid scooter's energy consumption through real urban drive cycle using imu. *Vehicles*, 4(1):297–313, 2022.
- [23] Bruno Korbar, Du Tran, and Lorenzo Torresani. Cooperative learning of audio and video models from self-supervised synchronization. *Advances in Neural Information Processing Systems*, 31, 2018.
- [24] Wojtek Krzanowski. *Principles of multivariate analysis*. OUP Oxford, 2000.
- [25] Wei-Yu Lee, Ljubomir Jovanov, and Wilfried Philips. Cross-modality attention and multimodal fusion transformer for pedestrian detection. In *European Conference on Computer Vision*, pages 608–623. Springer, 2022.
- [26] Yi-Shan Lee, Wei-Cheng Tseng, Fu-En Wang, and Min Sun. Vmcm: video and music matching via cross-modality lifting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2060–2069, 2024.
- [27] Haopeng Li, Qihong Ke, Mingming Gong, and Tom Drummond. Progressive video summarization via multimodal self-supervised learning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5584–5593, 2023.
- [28] Tianhong Li, Lijie Fan, Yuan Yuan, and Dina Katabi. Unsupervised learning for human sensing using radio signals. In

- Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3288–3297, 2022.
- [29] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision*, pages 740–755. Springer, 2014.
 - [30] Dongxin Liu, Tianshi Wang, Shengzhong Liu, Ruijie Wang, Shuochao Yao, and Tarek Abdelzaher. Contrastive self-supervised representation learning for sensing signals from the time-frequency perspective. In *2021 International Conference on Computer Communications and Networks (ICCCN)*, pages 1–10. IEEE, 2021.
 - [31] Hansi Liu, Abrar Alali, Mohamed Ibrahim, Bryan Bo Cao, Nicholas Meegan, Hongyu Li, Marco Gruteser, Shubham Jain, Kristin Dana, Ashwin Ashok, et al. Vi-fi: Associating moving subjects across vision and wireless sensors.
 - [32] Hansi Liu, Abrar Alali, Mohamed Ibrahim, Hongyu Li, Marco Gruteser, Shubham Jain, Kristin Dana, Ashwin Ashok, Bin Cheng, and Hongsheng Lu. Lost and found! associating target persons in camera surveillance footage with smartphone identifiers. In *Proceedings of the 19th Annual International Conference on Mobile Systems, Applications, and Services*, pages 499–500, 2021.
 - [33] Hansi Liu, Hongsheng Lu, Kristin Dana, and Marco Gruteser. Vifi-loc: Multi-modal pedestrian localization using gan with camera-phone correspondences. In *Proceedings of the 25th International Conference on Multimodal Interaction*, pages 661–669, 2023.
 - [34] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021.
 - [35] Alliance W Wi-Fi CERTIFIED Location. Brings wi-fi indoor positioning capabilities. *Wi-Fi Alliance*. [Online]. Available: <https://wi-fi.org/news-events/newsroom/wi-fi-certified-locationbrings-wi-fi-indoor-positioning-capabilities>, 2017.
 - [36] Mohammad Malekzadeh, Richard G Clegg, Andrea Cavallaro, and Hamed Haddadi. Mobile sensor data anonymization. In *Proceedings of the international conference on internet of things design and implementation*, pages 49–58, 2019.
 - [37] Pranay Manocha, Zeyu Jin, Richard Zhang, and Adam Finkelstein. Cdpam: Contrastive learning for perceptual audio similarity. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 196–200. IEEE, 2021.
 - [38] Alessandro Masullo, Tilo Burghardt, Dima Damen, Toby Perrett, and Majid Mirmehdi. Who goes there? exploiting silhouettes and wearable signals for subject identification in multi-person environments. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 0–0, 2019.
 - [39] Alessandro Masullo, Tilo Burghardt, Dima Damen, Toby Perrett, and Majid Mirmehdi. Person re-id by fusion of video silhouettes and wearable signals for home monitoring applications. *Sensors*, 20(9):2576, 2020.
 - [40] Gledson Melotti, Cristiano Premebida, Nuno MM da S Gonçalves, Urbano JC Nunes, and Diego R Faria. Multi-modal cnn pedestrian classification: a study on combining lidar and camera data. In *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, pages 3138–3143. IEEE, 2018.
 - [41] Anton Milan, Laura Leal-Taixé, Ian Reid, Stefan Roth, and Konrad Schindler. Mot16: A benchmark for multi-object tracking. *arXiv preprint arXiv:1603.00831*, 2016.
 - [42] Shentong Mo, Zhun Sun, and Chao Li. Multi-level contrastive learning for self-supervised vision transformers. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2778–2787, 2023.
 - [43] Namuk Park and Songkuk Kim. How do vision transformers work? *arXiv preprint arXiv:2202.06709*, 2022.
 - [44] Andra Petrovai and Sergiu Nedevschi. Monodvps: A self-supervised monocular depth estimation approach to depth-aware video panoptic segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 3077–3086, 2023.
 - [45] Sanka Rasnayaka and Terence Sim. Your tattletale gait privacy invasiveness of imu gait data. In *2020 IEEE International Joint Conference on Biometrics (IJCB)*, pages 1–10. IEEE, 2020.
 - [46] Jorge-L Reyes-Ortiz, Luca Oneto, Albert Samà, Xavier Parra, and Davide Anguita. Transition-aware human activity recognition using smartphones. *Neurocomputing*, 171: 754–767, 2016.
 - [47] Ergys Ristani, Francesco Solera, Roger Zou, Rita Cucchiara, and Carlo Tomasi. Performance measures and a data set for multi-target, multi-camera tracking. In *European conference on computer vision*, pages 17–35. Springer, 2016.
 - [48] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision (IJCV)*, 115 (3):211–252, 2015.
 - [49] Sara Sarto, Manuele Barraco, Marcella Cornia, Lorenzo Baraldi, and Rita Cucchiara. Positive-augmented contrastive learning for image and video captioning evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6914–6924, 2023.
 - [50] Muhammad Shoaib, Stephan Bosch, Ozlem Durmaz Incel, Hans Scholten, and Paul JM Havinga. Fusion of smartphone motion sensors for physical activity recognition. *Sensors*, 14 (6):10146–10176, 2014.
 - [51] Luciano Spinello, Rudolph Triebel, and Roland Siegwart. Multimodal detection and tracking of pedestrians in urban environments with explicit ground plane extraction. In *2008 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1823–1829. IEEE, 2008.
 - [52] Allan Stisen, Henrik Blunck, Sourav Bhattacharya, Thor Siger Prentow, Mikkel Baun Kjærgaard, Anind Dey, Tobias Sonne, and Mads Møller Jensen. Smart devices are different: Assessing and mitigating mobile sensing heterogeneities

- for activity recognition. In *Proceedings of the 13th ACM conference on embedded networked sensor systems*, pages 127–140, 2015.
- [53] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Herve Jegou. Training data-efficient image transformers & distillation through attention. In *International Conference on Machine Learning*, pages 10347–10357, 2021.
- [54] Boyuan Wang, Xuelin Liu, Baoguo Yu, Ruicai Jia, and Xingli Gan. Pedestrian dead reckoning based on motion mode recognition using a smartphone. *Sensors*, 18(6):1811, 2018.
- [55] He Wang, Xuan Bao, Romit Roy Choudhury, and Srihari Nelakuditi. Visually fingerprinting humans without face recognition. In *Proceedings of the 13th Annual International Conference on Mobile Systems, Applications, and Services*, pages 345–358, 2015.
- [56] Chen Wei, Haoqi Fan, Saining Xie, Chao-Yuan Wu, Alan Yuille, and Christoph Feichtenhofer. Masked feature prediction for self-supervised visual pre-training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14668–14678, 2022.
- [57] Huatao Xu, Pengfei Zhou, Rui Tan, Mo Li, and Guobin Shen. Limu-bert: Unleashing the potential of unlabeled data for imu sensing applications. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, pages 220–233, 2021.
- [58] Ke Xu, Jiangtao Wang, Le Zhang, Hongyuan Zhu, and Dingchang Zheng. Dual-stream contrastive learning for channel state information based human activity recognition. *IEEE Journal of Biomedical and Health Informatics*, 27(1): 329–338, 2022.
- [59] Jianfei Yang, Xinyan Chen, Han Zou, Dazhuo Wang, and Lihua Xie. Autofi: Toward automatic wi-fi human sensing via geometric self-supervised learning. *IEEE Internet of Things Journal*, 10(8):7416–7425, 2022.
- [60] Haichao Zhang, Yi Xu, Hongsheng Lu, Takayuki Shimizu, and Yun Fu. Layout sequence prediction from noisy mobile modality. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 3965–3974, 2023.
- [61] Haichao Zhang, Yi Xu, Hongsheng Lu, Takayuki Shimizu, and Yun Fu. Oostraj: Out-of-sight trajectory prediction with vision-positioning denoising. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14802–14811, 2024.
- [62] Haichao Zhang, Yi Xu, and Yun Fu. Out-of-sight trajectories: Tracking, fusion, and prediction. *arXiv preprint arXiv:2509.15219*, 2025.
- [63] Wenru Zheng, Ryota Yoshihashi, Rei Kawakami, Ikuro Sato, and Asako Kanezaki. Multi event localization by audio-visual fusion with omnidirectional camera and microphone array. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2566–2574, 2023.
- [64] Han Zou, Jianfei Yang, Hari Prasanna Das, Huihan Liu, Yuxun Zhou, and Costas J Spanos. Wifi and vision multimodal learning for accurate and robust device-free human activity recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, pages 0–0, 2019.